

LESSON 14.8

MAKING PREDICTIONS



LESSON INTRODUCTION

MA.8.DP.2.1 Determine the sample space for a repeated experiment.

MA.8.DP.2.2 Find the theoretical probability of an event in relation to a repeated experiment.

MA.8.DP.2.3 Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.

LEARNING TARGETS

- I can use theoretical probability to make predictions.

BENCHMARK COHERENCE

BUILDING ON

MA.7.DP.2.1 Determine the sample space for a simple experiment.

MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.

MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.

WORKING TOWARD

MA.912.DP.4.1 Given a mathematical or real-world situation, calculate the appropriate permutation or combination.

MA.912.DP.4.3 Calculate the conditional probability of two events and interpret the result in terms of its context.

MA.912.DP.3.5 Solve real-world problems involving univariate and bivariate categorical data.

ESSENTIAL QUESTIONS

- How can probabilities be used to make predictions?

LESSON NARRATIVE

This lesson opens with a review of adding rational numbers and with determining whether or not the given rational numbers could be a probability distribution, allowing students to apply knowledge from earlier lessons. Students collaboratively and independently practice finding probabilities and making predictions from real-world examples throughout the remaining components. The lesson guides students through making predictions from repeated experiments.

COMPONENT SUMMARY

14.8.1 WARM-UP (~5 MIN.)

Determine whether given probabilities could be in a probability distribution

14.8.3 GUIDED PRACTICE (10 MIN.)

Probability and predictions from a repeated experiment

14.8.2 COLLABORATION (10 MIN.)

Use probability to make predictions

14.8.4 YOUR TURN! (10-15 MIN.)

Probability and predictions involving repeated experiments and real-world scenarios

14.8.5 WRAP-UP (~5 MIN.)

Self-reflection on the content of the lesson

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Probability distribution
- Replacement

MATERIALS

- (Optional) Spinner

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse the value of the favorable outcomes with the value of the total outcomes when calculating probability from real-life situations.
- Students may assume replacement changes the probability.

THINGS TO CONSIDER

- This lesson brings together a lot of previously learned content. Consider posting definitions, examples, and anchor charts created throughout the unit to assist students with accessing prior knowledge.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

14.8.3 Guided Practice provides an opportunity to use the Collect and Display routine. After students have determined whether or not they would play the game and their reasoning, collect the answers and explanations. Display the collected information for the class to view and reflect on.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.7.1 Real-World Contexts**

Mathematics in Real-World Contexts covers relating mathematical concepts to real-life situations. By providing students with various real-world circumstances, this lesson allows students to experience repeated experiments in real-life situations.

DIFFERENTIATE INSTRUCTION

This lesson provides an opportunity to engage students through a variety of visual and kinesthetic entry points. Consider allowing students to work with a spinner (digital or real) and coin to provide a kinesthetic entry point during 14.8.2 Collaboration. Students can spin the spinner and flip the coin to demonstrate the probability of different events. This can also provide an additional means of engagement.

Additional differentiation opportunities are denoted throughout the Lesson Guide.

14.8.1 WARM-UP: SUM PROBABILITIES**DIFFERENTIATE INSTRUCTION**

Consider allowing students to use a calculator for question 1. The purpose of the Warm-Up is to determine their understanding of question 2 more than their ability to add the probabilities.

**14.8.1 Warm-Up: Sum Probabilities**

- Find the sum of the following numbers.

$$\frac{1}{5} \quad 0.008 \quad 33\frac{1}{3}\% \quad 15\% \quad \frac{31}{500}$$

0.75333 ...

- Explain whether or not these numbers could be the probabilities in a probability distribution.
They could not be probabilities in a distribution because they do not total to 1.



14.8.2 Collaboration: Solving Problems with Probability

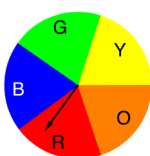
- Three red marbles, 3 blue marbles, 3 yellow marbles, and 3 orange marbles are in a bag. A representation of the marbles is shown.



- If a marble is chosen 20 times, with replacement, how many times do you expect to choose a blue marble?
5 times
- Suppose two marbles are chosen, with replacement. What is the probability that a blue marble is chosen both times?
0.0625 or 6.25%
- If this experiment is repeated 40 times (for a total of 80 marbles), how many times do you expect to choose two blue marbles?
2.5; 2 – 3 times

- As part of a game, Spinner B is spun and a coin is flipped.

Spinner B



Coin



If the spinner lands on a primary color and the coin lands on heads, the player wins.

- What is the probability of winning the game?
 $(0.6)(0.5) = 0.3$ or 30%
- If the game is played 250 times, what is the predicted number of wins?
75 times



14.8.3 Guided Practice: Predictions in Repeated Experiments

Suppose a game is played where a fair coin is tossed 5 times.

- If none of the tosses result in heads or all of the tosses result in heads, the player wins.
- In all other outcomes, the player loses.

The probability distribution for the possible number of heads is shown below.

Number of Heads	0	1	2	3	4	5
Probability	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

- What is the probability that a player wins on their first game?
6.25 %
- What is the probability that a player loses on their first game?
93.75%
- If a player plays this game 64 times, what is the predicted number of wins?
4 wins
- How many times will a player need to play the game in order to win 10 times?
160 times
- If the winning prize is \$20 and it costs \$5 to play the game, would you want to play this game? Explain your reasoning.
Answers will vary. Not likely. The probability of winning is low at only $\frac{2}{32}$. Theoretically, it would take 16 games to win once, which costs \$80 ($\5×16) to win \$20.

14.8.2 COLLABORATION: SOLVING PROBLEMS WITH PROBABILITY

ENRICHMENT

In order to solve question 1a, students must understand how replacing a marble affects or does not affect the probability. Ask students questions to strengthen this understanding, such as:

- What are other common words that mean the same as “replacement”?
- If the first marble is chosen and replaced, would the probability of the second marble change? Why or why not?

14.8.3 GUIDED PRACTICE: PREDICTIONS IN REPEATED EXPERIMENTS

DIFFERENTIATE INSTRUCTION

Allow informal observation throughout the first two activities to dictate how “guided” this Guided Practice should be for you and your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group.

DIFFERENTIATE INSTRUCTION

Consider providing sentence stems to assist students with clearly communicating their explanations for question 5, such as:

- The cost and prize money to play this game makes me want to play the game because_____.
- The cost and prize money to play this game does not make me want to play the game because_____.
- The money and chances of winning make me _____ to play this game since_____.

14.8.4 YOUR TURN!



14.8.4 Your Turn!

- Nikita decides to buy some chicken eggs to hatch. She buys 3 eggs because she wants to have three chickens in her yard. She finds out after she orders the eggs that there is a 50% chance each egg will hatch.
 - Find the theoretical probability that only one of the eggs will hatch.
 $P(\text{exactly one egg hatches}) = 0.375$
 - Find the theoretical probability that exactly 2 of the eggs will hatch.
 $P(\text{exactly 2 eggs hatch}) = 0.375$
 - How likely is it that all three of Nikita's eggs hatch?
 $P(\text{all 3 eggs hatch}) = 0.125$
 - If she wants to have three chickens, explain how many eggs she should buy.
Answers will vary. She should buy more than 3 eggs if she wants to have three chickens in her yard. If she buys only 3 eggs, there is only a 12.5% chance that all three will hatch. The likelihood of getting 3 chickens will increase if she has more eggs to start.
- Marianne Douglass has a 26-card deck of cards, each with a unique letter of the alphabet on it.
 - If she chooses two cards, with replacement, what is the probability she chooses her initials, MD, in the correct order?
 $1/676 = 0.00148$
 - If she chooses a card 300 times, how many outcomes including an A are expected?
 $11.54 \text{ times or } \sim 12 \text{ times}$

ENRICHMENT

To confirm that students understand the probability and predictions involved for question 3, ask students questions such as:

- How did Ann use the theoretical probability to predict her experimental results?
- Would the theoretical probability change if Ann spun the spinner 35 times in her experiment?

- Ann spins a fair spinner sectioned into seven equal parts, one for each day of the week, 350 times. She correctly predicts that she will land on Monday 50 times because the theoretical probability that she lands on Monday is

$$\frac{1}{7}$$

- A fair six-sided die is rolled four times. The probability of rolling the same number on all four rolls is $\frac{1}{216}$. If this experiment is performed 1000 times (for a total of 4000 rolls), predict the number of repetitions that will result in all four rolls landing on the same number.
 $4.63 \text{ times or } \sim 5 \text{ times}$



14.8.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.



I need help.



I got it.



I could help someone.

Determining the sample space for a repeated experiment.	  
Finding the theoretical probability of an event related to a repeated experiment.	  
Solving problems using probability.	  
Making predictions based on theoretical probability.	  

Answers will vary.

14.8.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider using student responses from prior self-reflections to consider a student’s learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.